Proposal for Second Year Paper

Ke Jiang

1 Introduction

1.1 Motivation

U.S. cities, on average, have less transit infrastructure. Freemark, 2023 reports that although transit length and ridership are growing globally, the United States appears to have stagnated since the 1980s. Putting aside China, which accounts for 40% of total metro length, by 2023 the total metro length of the United States had been surpassed by India and was nearly half that of the EU (fig. 1). Even when incorporating light rail or streetcars, on which U.S. cities place more emphasis, transit construction has declined since the 1970s and has struggled to keep up with population growth. The reasons behind this pattern are puzzling but nontrivial. Transportation infrastructure is often endogenous: it serves as a tool to boost economic development by reducing spatial frictions, but it can also be distorted (Donaldson, 2025). In particular, policymakers may intentionally build infrastructure to boost economic activity, or they may choose not to build if it is too expensive relative to expected returns. In general, they evaluate the returns and costs of new infrastructure before making investment decisions. Standard evaluations of the returns to transportation infrastructure involve measuring travel time savings or distributional effects (Fogel, 1964; Tsivanidis, 2026; Severen, 2021). Both Tsivanidis (2026) and Severen (2021) find that the returns to new transit infrastructure are lower than expected, and Severen (2021) even finds negative returns, due to limited housing supply near stations. Severen (2021) also points to dispersed workplaces and a polycentric city structure as another reason for the low return to transit. In summary, restrictive zoning can limit the housing supply response to transit investment, which in turn can discourage transit investment.

However, the literature estimating transportation infrastructure often takes these housing-supply restrictions as given, whereas in reality, aside from certain geographic constraints, such restrictions are often the result of local zoning policies. Zoning is often politically supported by incumbent residents because it protects neighborhood quality and property values. In particular, restrictive zoning can limit housing supply, which protects local property values, reduces congestion in local public goods and amenities, and preserves environmental quality and neighborhood character (Rollet and Weiwu, 2025). However, these local benefits may come at broader citywide costs. In particular, restrictive zoning reduces the elasticity of housing supply in desirable locations (Glaeser and Gyourko, 2002), and land-use regulation pushes forward job-housing separation, raises commuting costs, and contributes to urban sprawl (Glaeser and Kahn, 2003). The resulting low-density city layout can also lower the economic return to public transit and discourage investment in public transit. Lower transit investment can lead to more car-oriented commuting, which further decentralizes the city (Baum-Snow, 2007). Such decentralization and car dominance, in turn, reinforce the demand for low-density cities and therefore for zoning regulations. This self-enforcing pattern can be a key driver of urban sprawl and may help explain the stagnation of transit infrastructure in U.S. cities.

This self-reinforcing pattern between restrictive zoning and low transit investment can be particularly problematic in the context of U.S. cities. On the one hand, low-density sprawl requires more road infrastructure per capita, and the construction and maintenance costs may not be sustainable (Brueckner, 2000). On the other hand, car dependence and long commuting distances can increase congestion costs that are not internalized by households. Moreover, low transit investment can also limit low-income households' access to job opportunities, which can exacerbate economic inequality (Glaeser, Kahn, and Rappaport, 2008).

In summary, the discussion points to a broader but potentially underexplored problem: the full effect of zoning may be larger than we currently understand if we consider its effect on endogenous transit investment decisions. In particular, this proposal aims to explore the following questions: First, how does local zoning affect metropolitan transportation investment decisions? Second, what are the welfare implications of this interaction, and how much inefficiency can be mitigated compared to a socially optimal benchmark in which zoning and transportation are coordinated?

1.2 Literature Review

This proposal draws on three interconnected bodies of literature: studies on transportation infrastructure and its evaluation, literature on zoning and land-use regulation, and research on urban sprawl and city growth patterns.

1.2.1 Transportation Infrastructure, Optimal Structure, and Evaluation

A large body of literature examines how transportation infrastructure shapes spatial economic patterns and how to evaluate infrastructure investments. Donaldson (2025) provides a comprehensive framework for evaluating transportation infrastructure impacts, emphasizing the importance of general equilibrium effects and showing how infrastructure changes affect economic activity through multiple channels. The quantitative spatial economics framework is essential for this evaluation: Redding (2025) surveys quantitative urban economics methods, while Redding and Rossi-Hansberg (2017) details the development of quantitative spatial models that can integrate multiple dimensions of infrastructure impacts. Contemporary applications of these methods highlight that traditional benefit-cost analyses may not capture full infrastructure impacts. Severen (2026) describes how quantitative spatial models can evaluate transportation improvements within cities, providing practical guidance for implementation. Importantly, Fajgelbaum et al. (2021) develops a framework integrating spatial epidemiology and commuting networks, demonstrating the importance of accounting for spatial structure in policy design. The returns to transit infrastructure specifically have received attention. Tsivanidis (2026) evaluates Bogotá's TransMilenio system and finds that transit returns depend critically on complementary policies affecting land use. Baum-Snow and Kahn (2000) studies rail transit expansion in five U.S. cities and finds modest impacts on usage and housing values, suggesting limitations to transit investment absent other changes. Kahn (2007) documents significant heterogeneity in transit effects across 14 cities, with gentrification outcomes varying substantially based on local conditions like station type (walk-and-ride versus park-and-ride).

The interaction between transportation infrastructure and congestion is increasingly recognized. Akbar et al. (2023) analyzes the relationship between mobility and congestion in urban India, showing how congestion affects the effectiveness of transportation improvements. Allen and Arkolakis (2022) develops a general equilibrium framework with endogenous congestion to evaluate welfare impacts of transportation improvements, finding substantial variation in returns across different infrastructure investments. Kashner and Ross (2025) relates congestion to upzoning, showing that the effect of upzoning on housing supply depends on where upzoning occurs. Specifically, upzoning near transit stations has lower congestion effects than upzoning in other areas. Integrating congestion into the analysis of both zoning and transportation is critical for my proposal, which I believe may be a key channel explaining welfare differences. My proposal contributes by modeling the endogeneity of zoning regulations and transportation investment, which are often taken as given in the literature.

1.2.2 Zoning and Land-Use Regulation

Zoning and land-use regulations have emerged as critical determinants of urban form and economic efficiency. The foundational work of Glaeser and Gyourko (2002) demonstrates that zoning restrictions significantly reduce housing supply elasticity, with particularly severe constraints in high-demand cities. This finding has profound implications: Hsieh and Moretti (2019) quantifies the aggregate costs of housing constraints, estimating that zoning restrictions lowered U.S. aggregate growth by 36 percent between 1964 and 2009 by preventing spatial reallocation of labor to high-productivity locations.

Recent work has continued this examination with new evidence and methods. Rollet and Weiwu (2025) measures winners and losers from increasing housing supply, highlighting the distributional tensions inherent in zoning reform—local residents benefit from restrictive zoning even when it imposes broader costs. Song (2025) estimates the effects of residential zoning on U.S. housing markets, providing detailed evidence of how zoning shapes housing supply responses. Mast (2024) uses electoral reforms as a source of exogenous variation in local control and finds that decentralization of land-use authority reduces housing permits by 20 percent, with larger effects in affluent areas. These results highlight how local political economy affects housing supply. Importantly, Baum-Snow and Duranton (2025) surveys housing supply and affordability, providing context for understanding how zoning interacts with other factors affecting housing markets. A critical insight from recent work is that land-use regulations interact with transportation investment. Hu and Wang (2025) studies how subway expansion affects outcomes under different zoning regimes, finding that rigid land-use regulation limits the benefits of transit infrastructure. My proposal emphasizes that restrictive zoning can also discourage transit investment, which is a key channel that has not been fully explored in the literature.

1.2.3 Urban Sprawl and City Growth

Urban sprawl represents a fundamental concern in metropolitan economics. Brueckner (2000) provides the canonical analysis of sprawl causes and remedies, identifying three market failures: incomplete internalization of open-space benefits, failure to charge commuters for congestion externalities, and insufficient cost-recovery for infrastructure. Baum-Snow (2007), discussed above, documents empirically how highways have driven suburban decentralization, with implications for sprawl patterns. Glaeser and Kahn (2003) documents that sprawl is ubiquitous and continuing to expand, but is not caused by bad urban planning; rather, it reflects increasing demand for car-based living, though low-income people who cannot afford cars are left behind. My proposal seeks to explain another mechanism where car ownership may not merely be a preference but also an equilibrium outcome.

The literature collectively demonstrates that sprawl outcomes reflect a complex interaction between fundamental forces, such as population growth, rising incomes, falling transport costs, and structural changes. Cuberes, Desmet, and Rappaport (2021) documents urban growth shadows and access effects, showing how proximity to large metropolitan areas affects growth patterns through both competition and market-access mechanisms. Desmet and Rappaport (2017) traces the long-term evolution of U.S. settlement patterns from 1800-2000, documenting the transition toward increasingly dispersed city systems. Structural change and land use are intimately connected. Coeurdacier, Oswald, and Teignier (2022) studies how structural change in the economy drives land-use changes and urban expansion. International evidence adds important insights: Zheng and Kahn (2013) documents how China's bullet trains facilitate market integration and mitigate megacity growth costs by allowing workers to access large labor markets from secondary cities. Building on these integrated insights, my proposal aims to explore how the interaction between zoning and transportation investment shapes sprawl patterns, and how more coordinated policies could mitigate sprawl and its associated costs.

1.3 Planned Empirical Work

The empirical part of the project will examine whether metropolitan transportation investment is shaped by the expected economic returns to transit. I plan to focus on three related hypotheses:

1. Transportation investment decisions respond to the expected costs and benefits of new infrastructure.

2. Older cities that developed extensive transit systems before widespread automobile dominance have higher returns to transit investment and therefore maintain larger transit networks, whereas newer, more car-oriented cities face lower returns.

3. Restrictive zoning lowers the return to transit investment by limiting the density and ridership response around high-access locations.

The main data sources will include the National Transit Database (NTD) and the Transit Explorer database, which provide information on transit volumes, revenues, investment, and opening dates. I will measure zoning constraints using housing supply elasticity estimates from Baum-Snow and Han (2024).

2 Model

Following Severen (2026), I aim to develop a quantitative spatial model to answer the questions raised in the introduction. The model will capture the key features of metropolitan spatial structure, including heterogeneous locations, commuting patterns, and land-use decisions. It will also incorporate the institutional structure of local zoning and metropolitan transportation investment.

For now, I am considering a model in which municipalities and the transportation authority make decisions sequentially, with municipalities choosing zoning first and the transportation authority responding with infrastructure investment. This captures the idea that zoning decisions are often more entrenched and harder to change than transportation investments, which can be adjusted more frequently. However, I am open to alternative specifications in which the timing of decisions may differ.

2.1 Quantitative Spatial Model

2.1.1 Setup

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, N\}$. These locations are partitioned into a set of municipalities $g \in G$, such that

$$S = \bigcup_{g \in G} S^g, \qquad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed amount of land $\bar{H}_i$, which can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

Land use is further subject to zoning regulations. Let Z_i^R and Z_i^F denote the residential and commercial zoning restrictions at location i. These regulations impose upper bounds on feasible development:

$$H_i^R \leq \bar{H}_i^R(Z_i), \qquad H_i^F \leq \bar{H}_i^F(Z_i),$$

where stricter zoning lowers the feasible density of development.

Locations are connected through a transportation network. For any pair of residence and workplace locations (i, j), households can choose a commuting route $r \in R_{ij}$. Following **Fajgelbaum2020-yv**, the generalized commuting cost associated with route r is denoted by τ_{ijr}, which depends on transportation infrastructure and congestion:

$$\tau_{ijr} = \tau_{ijr}(I^R, I^T, \mathbf{n}),$$

where I^R denotes metropolitan road investment, I^T denotes metropolitan public transit investment, and $\mathbf{n}$ collects equilibrium traffic and transit usage across routes, which generate congestion. Road investment primarily lowers the cost of private commuting, whereas transit investment lowers the cost of public commuting and is more valuable when land around high-access locations can be developed more intensively.

There are three types of agents in the model: households, firms, and governments. Households choose where to live, where to work, and how to commute. Competitive firms choose labor and commercial land to produce a freely tradable good. Government decisions are made at two levels. Municipal governments choose local zoning regulations, while a metropolitan transportation authority chooses transportation investment for the urban area as a whole.

2.1.2 Households

A household chooses a residence location i, a workplace location j, and a commuting route $r \in R_{ij}$. Conditional on these choices, the household consumes a composite tradable good Q_{ij} and residential land h^R_{ij}. Preferences are given by

$$u_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \left(\frac{Q_{ij}}{\beta}\right)^{\beta} \left(\frac{h^R_{ij}}{1-\beta}\right)^{1-\beta} v_{ijr}(w),$$

where B_i is the residential amenity at location i, $\beta \in (0,1)$, and $v_{ijr}(w)$ is an idiosyncratic preference shock.

Let q^R_i denote the rental price of residential land at location i, and w_j the wage earned at workplace location j. The household's budget constraint is

$$Q_{ij} + q^R_i h^R_{ij} = w_j.$$

Solving the within-choice consumption problem yields the indirect utility

$$V_{ijr}(w) = \frac{B_i}{\tau_{ijr}} \frac{w_j}{(q^R_i)^{1-\beta}} v_{ijr}(w),$$

up to a constant common to all choices.

To allow for a nested discrete-choice structure, let the idiosyncratic shocks follow a generalized extreme value distribution. The lower nest corresponds to route choice conditional on a residential-workplace pair (i,j), while the upper nest corresponds to the joint choice of residence and workplace. Formally,

$$G(\{v_{ijr}(w)\}) = \exp\left(-\sum_{i,j \in S^2} \left(\sum_{r \in R_{ij}} v_{ijr}^{-1/\rho}\right)^{\rho/\theta}\right),$$

where ρ and θ govern substitution within and across nests.

Conditional on (i,j), households choose the commuting route that maximizes indirect utility. At the upper level, they choose residence and workplace locations based on the expected value from the lower nest. Thus, household sorting across space depends on wages, rents, local amenities, commuting costs, and transportation infrastructure.

2.1.3 Firms

At each workplace location j, competitive firms produce a freely tradable good using effective labor and commercial land. Let output be given by

$$Y_j = A_j F(L_j, H^F_j),$$

where A_j is location-specific productivity, L_j is labor employed at location j, and H^F_j is commercial land use. Firms solve

$$\max_{L_j, H^F_j} \; A_j F(L_j, H^F_j) - w_j L_j - q^F_j H^F_j,$$

where q^F_j denotes the rental price of commercial land.

Profit maximization implies standard labor-demand and land-demand conditions. In equilibrium, wages and commercial rents adjust so that labor and land markets clear.

2.1.4 Municipal Zoning Decisions

Each municipality $g \in G$ chooses a vector of zoning policies

$$Z^g = \{Z_i\}_{i \in S^g}.$$

To avoid modeling an explicit dynamic voting game, I assume municipal governments act as reduced-form representatives of incumbent residents. This captures the idea that local land-use policy is shaped by the interests of existing residents, who value neighborhood amenities, lower congestion, and higher property values, but do not internalize metropolitan-wide effects on housing supply, commuting, or transit efficiency.

Accordingly, municipality g solves

$$\max_{Z^g} \; W^g(Z^g; I^R, I^T, Z^{-g}),$$

where Z^{-g} denotes the zoning choices of other municipalities, and W^g is incumbent welfare in municipality g. In reduced form, this objective can be written as

$$W^g = \sum_{i \in S^g} \left[\omega_1 P_i + \omega_2 B_i^{\text{loc}} - \omega_3 N_i\right],$$

where P_i captures local property-value considerations, B_i^{loc} captures neighborhood amenities, and N_i captures congestion or crowding costs associated with additional development. Because this objective is local, municipalities do not internalize the effect of zoning restrictions on regionwide housing allocation, commuting patterns, or the return to regional transit investment.

2.1.5 Metropolitan Transportation Authority

A metropolitan transportation authority chooses investment in road and transit infrastructure for the region as a whole. Let its policy vector be

$$I = (I^R, I^T).$$

Taking local zoning as given, the authority solves

$$\max_{I^R, I^T} \; W^M(I^R, I^T; Z) - C^R(I^R) - C^T(I^T),$$

where $W^M(\cdot)$ denotes the regional benefit from transportation infrastructure, Z is the vector of municipal zoning policies, and $C^R(\cdot)$, $C^T(\cdot)$ are the costs of road and transit investment.

The distinction between road and transit investment is primarily in how their returns are evaluated. Road investment primarily reduces the cost of private commuting, and its returns depend on the overall level of car-based commuting. Transit investment reduces the cost of public commuting, and its returns depend critically on the density and ridership response to transit-accessible locations, which in turn depends on local zoning.

The key institutional friction is that the metropolitan authority controls transportation investment but does not control zoning. Since local land use determines whether high-access locations can absorb additional residential and commercial development, the marginal return to transit investment depends on municipal zoning decisions. In particular, restrictive zoning near transit-accessible locations weakens the density and ridership response to public transit investment.

2.1.6 Equilibrium

A decentralized equilibrium consists of

$$\{q_i^R, q_i^F, w_j, H_i^R, H_i^F, L_j, Z^g, I^R, I^T\}_{i,j\in S,\, g\in G}$$

such that:

1. Households choose residence, workplace, and commuting route to maximize utility.
2. Firms choose labor and commercial land to maximize profits.
3. Each municipality chooses zoning to maximize incumbent resident welfare.
4. The metropolitan transportation authority chooses road and transit investment to maximize regional net benefits, taking zoning as given.
5. Residential land, commercial land, and labor markets clear.

2.1.7 Inefficiency and the Source of Sprawl

The main distortion in the model arises from a mismatch in governance scales. Zoning is chosen locally by municipalities that represent incumbent interests, while transportation is chosen at the metropolitan level. Municipalities have incentives to restrict development in high-access locations in order to preserve local amenities and limit congestion. However, they do not internalize the broader metropolitan costs of such restrictions, including higher housing prices, greater job-housing separation, and lower returns to public transit investment.

As a result, restrictive zoning can prevent transportation improvements—especially transit investments—from translating into higher-density development near well-connected locations. Growth is instead pushed toward lower-cost, more distant areas, contributing to urban sprawl. In this sense, sprawl in the model is not simply a reflection of geography or household preferences, but may emerge as an inefficient equilibrium outcome of fragmented land-use regulation combined with metropolitan transportation investment.

2.1.8 Efficient Benchmark

To evaluate the inefficiency of the decentralized equilibrium, I compare it to a coordinated benchmark in which a planner chooses both transportation investment and zoning:

$$\max_{I^R, I^T, Z} W(I^R, I^T, Z) - C^R(I^R) - C^T(I^T),$$

where $W(\cdot)$ denotes aggregate metropolitan welfare. Unlike individual municipalities, the planner internalizes the effect of land-use restrictions on housing supply, commuting, and the productivity of transportation infrastructure.

Comparing the decentralized equilibrium with this benchmark highlights the extent to which fragmented zoning lowers the return to transit, biases investment toward road-oriented development, and generates excessive urban sprawl.

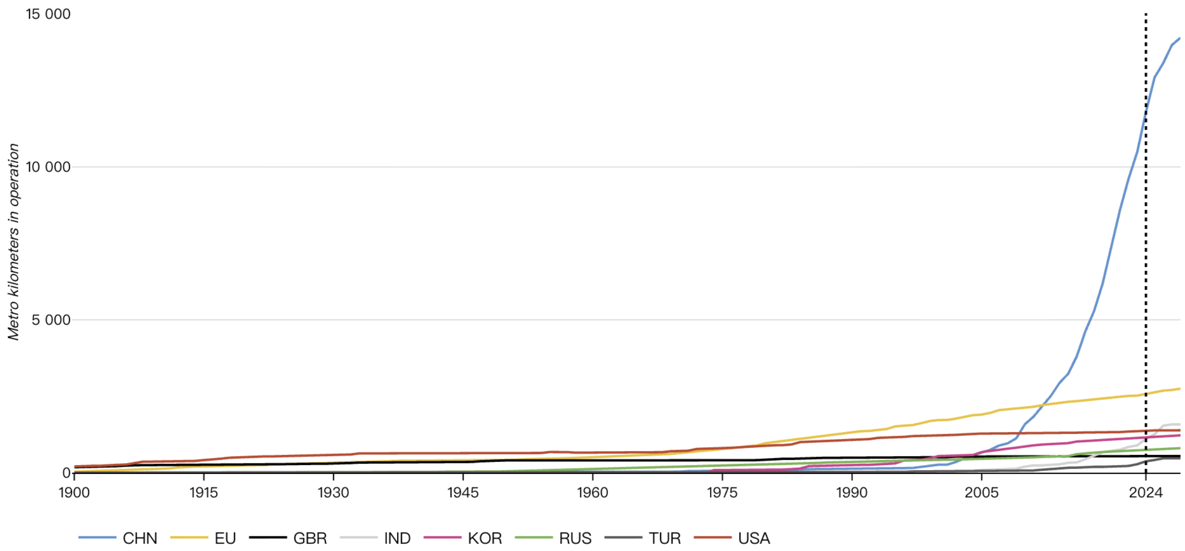

Figure 1: Metro kilometers in operation

References

[1] Prottoy Akbar et al. "Mobility and Congestion in Urban India". In: *American Economic Review* 113 (4 Apr. 2023), pp. 1083–1111. (Visited on 09/20/2023).

[2] Treb Allen and Costas Arkolakis. "The Welfare Effects of Transportation Infrastructure Improvements". en. In: *The Review of Economic Studies* 89 (6 Nov. 7, 2022), pp. 2911–2957.

[3] Nathaniel Baum-Snow. "Did Highways Cause Suburbanization?" In: *The Quarterly Journal of Economics* 122 (2 2007), pp. 775–805.

[4] Nathaniel Baum-Snow and Gilles Duranton. "Housing Supply and Housing Affordability". en. In: *Handbook of Regional and Urban Economics*. Vol. 6. Handbook of Regional and Urban Economics. Elsevier, 2025, pp. 353–461.

[5] Nathaniel Baum-Snow and Lu Han. "The microgeography of housing supply". en. In: *The Journal of Political Economy* 132 (6 June 1, 2024), pp. 1897–1946.

[6] Nathaniel Baum-Snow and Matthew E Kahn. "The Effects of New Public Projects to Expand Urban Rail Transit". en. In: *Journal of Public Economics* 77 (2 Aug. 2000), pp. 241–263.

[7] Jan K Brueckner. "Urban Sprawl: Diagnosis and Remedies". en. In: *International Regional Science Review* 23 (2 Apr. 2000), pp. 160–171.

[8] Nicolas Coeurdacier, Florian Oswald, and Marc Teignier. "Structural Change, Land Use, and Urban Expansion". In: *SSRN Electronic Journal* (2022).

[9] David Cuberes, Klaus Desmet, and Jordan Rappaport. "Urban Growth Shadows". In: *Journal of Urban Economics* 123 (May 1, 2021), p. 103334.

[10] Klaus Desmet and Jordan Rappaport. "The Settlement of the United States, 1800–2000: The Long Transition towards Gibrat's Law". en. In: *Journal of Urban Economics* 98 (Mar. 1, 2017), pp. 50–68. (Visited on 10/08/2025).

[11] Dave Donaldson. “Transport Infrastructure and Policy Evaluation”. en. In: *Handbook of Regional and Urban Economics*. Vol. 6. Handbook of Regional and Urban Economics. Elsevier, 2025, pp. 287–352.

[12] Pablo D Fajgelbaum et al. “Optimal Lockdown in a Commuting Network”. en. In: *American Economic Review: Insights* 3 (4 Dec. 1, 2021), pp. 503–522.

[13] Robert William Fogel. *Railroads and American Economic Growth*. Baltimore: Johns Hopkins Press., 1964.

[14] Yonah Freemark. *Once a Leader in Urban Rail Investment, the United States Now Trails*. en. June 7, 2023. (Visited on 04/28/2026).

[15] Edward L Glaeser and Joseph Gyourko. *The Impact of Zoning on Housing Affordability*. en. Research rep. 8835. Cambridge, MA, 2002.

[16] Edward L Glaeser and Matthew E Kahn. *Sprawl and Urban Growth*. en. Research rep. 9733. Cambridge, 2003.

[17] Edward L Glaeser, Matthew E Kahn, and Jordan Rappaport. “Why Do the Poor Live in Cities? The Role of Public Transportation”. en. In: *Journal of Urban Economics* 63 (1 Jan. 2008), pp. 1–24. DOI: 10.1016/j.jue.2006.12.004.

[18] Chang-Tai Hsieh and Enrico Moretti. “Housing Constraints and Spatial Misallocation”. en. In: *American Economic Journal: Macroeconomics* 11 (2 Apr. 1, 2019), pp. 1–39.

[19] Wenhao Hu and Lili Wang. “Building the Metropolis: Subway Expansion, Land Use Regulation, and Welfare”. en. In: *Journal of Housing Economics* 70 (102102 Dec. 2025), p. 102102.

[20] Matthew E Kahn. “Gentrification Trends in New Transit-Oriented Communities: Evidence from 14 Cities That Expanded and Built Rail Transit Systems”. en. In: *Real Estate Economics* 35 (2 June 2007), pp. 155–182.

[21] Zane Kashner and Brad Ross. *Measuring and Mitigating Traffic Externalities*. en. 2025.

[22] Evan Mast. “Warding off Development: Local Control, Housing Supply, and NIMBYs”. en. In: *The Review of Economics and Statistics* 106 (3 May 14, 2024), pp. 671–680.

[23] Stephen J Redding. “Quantitative Urban Economics”. en. In: *Handbook of Regional and Urban Economics*. Vol. 6. Handbook of Regional and Urban Economics. Elsevier, 2025, pp. 73–141.

[24] Stephen J Redding and Esteban Rossi-Hansberg. “Quantitative Spatial Economics”. en. In: *Annual Review of Economics* 9 (1 Aug. 2, 2017), pp. 21–58.

[25] Vincent Rollet and Laura Weiwu. *Measuring Winners and Losers from Increasing Housing Supply*. en. Research rep. 2025.

[26] Christopher Severen. “Commuting, Labor, and Housing Market Effects of Mass Transportation: Welfare and Identification”. en. In: *The Review of Economics and Statistics* 105 (5 Sept. 22, 2021), pp. 1–99.

[27] Christopher Severen. *Evaluating Transportation Improvements within Cities Using Quantitative Spatial Models*. en. Research rep. 2026.

[28] Jaehee Song. “The Effects of Residential Zoning in U.S. Housing Markets”. en. In: *Journal of Urban Economics* 149 (103784 Sept. 2025), p. 103784.

[29] Nick Tsivanidis. “Evaluating the Impact of Urban Transit Infrastructure: Evidence from Bogotá’s TransMilenio”. In: *American Economic Review* 116 (2 Feb. 2026), pp. 418–463. DOI: 10.1257/aer.20190874.

[30] Siqi Zheng and Matthew E Kahn. “China’s Bullet Trains Facilitate Market Integration and Mitigate the Cost of Megacity Growth”. In: *Proceedings of the National Academy of Sciences* 110 (14 2013), E1248–E1253.